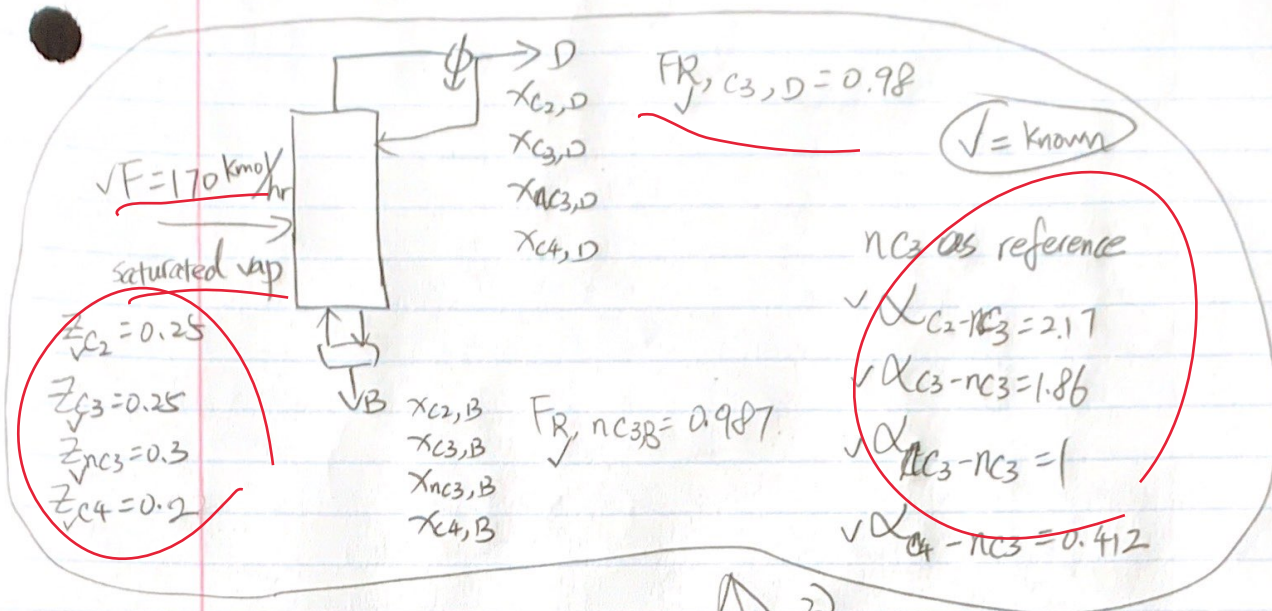




- 1) LNK = C_2 (ethanol)
 LK = i-propanol (C_3)
 HK = n-propanol (nC_3)
 HNK = n-butanol (C_4)

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2)

- 3) Assume all LNK are collected at the top
 HNK are collected at the bottom

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$$C_2: F \cdot Z_{C2} = D \cdot x_{D,C2} \quad | \quad B \cdot x_{B,C2} = 0$$

$$C_3: FR, C3, D \cdot F \cdot Z_{C3} = D \cdot x_{D,C3} \quad | \quad (1 - FR, C3, D) \cdot F \cdot Z_{C3} = B \cdot x_{B,C3}$$

$$nC_3: (1 - FR, nC3, B) \cdot F \cdot Z_{nC3} = D \cdot x_{D,nC3} \quad | \quad FR, nC3, B \cdot F \cdot Z_{nC3} = B \cdot x_{B,nC3}$$

$$C_4: 0 = D \cdot x_{D,C4} \quad | \quad F \cdot Z_{C4} = B \cdot x_{B,C4}$$

$$4) \quad \underline{D \cdot X_{D,c2}} = F \cdot Z_{c2} = 170 \frac{\text{kmol}}{\text{hr}} \cdot 0.25 = 42.5 \frac{\text{kmol}}{\text{hr}} \quad \checkmark$$

$$\underline{D \cdot X_{D,c3}} = FR_{c3,D} \cdot F \cdot Z_{c3} = 0.98 \times 170 \times 0.25 = 41.65 \frac{\text{kmol}}{\text{hr}} \quad \checkmark$$

$$\underline{D \cdot X_{D,nc3}} = (1 - FR_{nc3,B}) \cdot F \cdot Z_{nc3} = (1 - 0.987)(170)(0.3) = 0.663 \frac{\text{kmol}}{\text{hr}} \quad \checkmark$$

$$\underline{D \cdot X_{D,c4}} = 0 \quad \checkmark$$

$$\underline{D = \sum D \cdot X_{D,i}} = 42.5 + 41.65 + 0.663 = 84.813 \frac{\text{kmol}}{\text{hr}} \quad \text{(circled)}$$

$$\underline{X_{D,c2}} = \frac{D \cdot X_{D,c2}}{D} = \frac{42.5}{84.813} = \underline{0.5011}$$

$$\underline{X_{D,c3}} = \frac{D \cdot X_{D,c3}}{D} = \frac{41.65}{84.813} = \underline{0.49108}$$

$$\underline{X_{D,nc3}} = \frac{D \cdot X_{D,nc3}}{D} = \frac{0.663}{84.813} = \underline{0.0078172}$$

$$\underline{X_{D,c4}} = \frac{D \cdot X_{D,c4}}{D} = \underline{0}$$

$$5) \quad \underline{B \cdot X_{B,c2}} = 0$$

$$\underline{B \cdot X_{B,c3}} = (1 - FR_{c3,D}) \cdot F \cdot Z_{c3} = (1 - 0.98)(170)(0.25) = 0.85 \frac{\text{kmol}}{\text{hr}}$$

$$\underline{B \cdot X_{B,nc3}} = FR_{nc3,B} \cdot F \cdot Z_{nc3} = 0.987(170)(0.3) = 50.337 \frac{\text{kmol}}{\text{hr}}$$

$$\underline{B \cdot X_{B,c4}} = F \cdot Z_{c4} = 170 \cdot 0.2 = 34 \frac{\text{kmol}}{\text{hr}}$$

$$\underline{B = \sum B \cdot X_{B,i}} = 0.85 + 50.337 + 34 = 85.187 \frac{\text{kmol}}{\text{hr}}$$

- 1

5) continue

$$\underline{x_{B,C2} = \frac{0}{B} = 0}$$

$$\underline{x_{B,C3} = \frac{B \cdot x_{B,C3}}{B} = \frac{0.85}{85.187} = 0.009978}$$

$$\underline{x_{B,nc3} = \frac{B \cdot x_{B,nc3}}{B} = \frac{50.337}{85.187} = 0.5909}$$

$$\underline{x_{B,C4} = \frac{B \cdot x_{B,C4}}{B} = \frac{34}{85.187} = 0.399122}$$

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6) Find N_{min} & $N_{F,min}$

$$N_{min} = \frac{\ln \left[\left(\frac{FR_{D,C3}}{1 - FR_{B,nc3}} \right) / \left(\frac{1 - FR_{D,C3}}{FR_{B,nc3}} \right) \right]}{\ln \alpha_{c3-nc3}}$$

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$$N_{min} = \frac{\ln \left[\left(\frac{0.98}{1 - 0.987} \right) / \left(\frac{1 - 0.98}{0.987} \right) \right]}{\ln(1.86)} = \frac{9.27}{0.6205765} = 14.94$$

13.25
-1

$$N_{F,min} = \frac{\ln \left[\left(\frac{x_{LK,D}}{x_{HK,D}} \right) / \left(\frac{z_{LK}}{z_{HK}} \right) \right]}{\ln \alpha_{LK-HK}}$$

$$N_{F,min} = \frac{\ln \left[\left(\frac{0.49108}{0.0078172} \right) / \left(\frac{0.25}{0.3} \right) \right]}{\ln(1.86)} = \frac{4.32260217}{0.6205765} = 6.965462366$$

$$7) \Delta V_{\text{feed}} = F(1-q) \quad q = 0 \text{ b/c saturated vap}$$

$$\therefore \Delta V_{\text{feed}} = F(1-0) = F = 170 \text{ kmol/hr}$$

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$$8) \text{ Find } V_{\min}, \left(\frac{L}{D}\right)_{\min}, \phi = 1.67$$

$$V_{\min} = \sum \frac{\alpha_{ir} (D) X_{i,D,\text{dist}}}{\alpha_{ir} - \phi}$$

$$= \frac{\alpha_{C2-nc3} \cdot D \cdot X_{C2,D}}{\alpha_{C2-nc3} - \phi} + \frac{\alpha_{C3-nc3} \cdot D \cdot X_{C3,D}}{\alpha_{C3-nc3} - \phi}$$

$$+ \frac{\alpha_{nC3-nc3} \cdot D \cdot X_{nC3,D}}{\alpha_{nC3-nc3} - \phi} + \frac{\alpha_{C4-nc3} \cdot D \cdot X_{C4,D}}{\alpha_{C4-nc3} - \phi}$$

$$= \frac{2.17 (42.5)}{2.17 - 1.67} + \frac{1.86 (41.65)}{1.86 - 1.67} + \frac{1 \cdot (0.663)}{1 - 1.67} + \frac{0.412 (0)}{0.412 - 1.67}$$

$$= 184.45 + 407.7315789 + (-0.9895522388)$$

$$V_{\min} = 591.1920267 \text{ (kmol/hr)} \quad \checkmark$$

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$$V = D + L$$

$$L_{\min} = V_{\min} - (D)_{\min} = (591.1920267 - 84.813) \text{ kmol/hr}$$

$$= 506.3790267 \text{ kmol/hr}$$

$$\left(\frac{L}{D}\right)_{\min} = \frac{506.3790267}{84.813} = 5.970535492 \quad \checkmark$$

9) If $\frac{L}{D} = 1.1 \left(\frac{L}{D}\right)_{\min}$ find N & N_F

$$\frac{L}{D} = 1.1 (5.97053542) = 6.567589041$$

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$$\frac{\frac{L}{D} - \left(\frac{L}{D}\right)_{\min}}{\frac{L}{D} + 1} = \frac{0.597053549}{7.567589041} = 0.078896138 \Rightarrow X$$

$$Y = \frac{N - N_{\min}}{N + 1} = 0.545827 - 0.591422X + \frac{0.002743}{X}$$

$$= 0.545827 - 0.0466609117 + 0.0347672278$$

$$\frac{N - N_{\min}}{N + 1} = 0.5339333161$$

$$N - N_{\min} = 0.5339333161 (N + 1)$$

13.25

14.94

$$N - 0.5339333161N = 0.5339333161 + N_{\min}$$

$$N = \frac{15.47393332}{(1 - 0.5339333161)} = 33.2011446$$

29.58

$$\frac{N_F}{N} = \frac{N_{F,\min}}{N_{\min}}$$

$$N_F = \frac{N_{F,\min}}{N_{\min}} \cdot N = \frac{6.965462366}{14.94} \cdot 33.2011446$$

15.63

$$N_F = 15.47932485$$